Agent tools on the server box

Monitoring, a CPU model, retrieval, web search and hardware health

RX 6700 XT box (Ubuntu, Ryzen 5 7600X) · served to agents over MCP · gemma4 12B as the answering model

October 2026 · reproducible with bench/rag.sh · results in docs/BENCHMARKS.md

Key indicators

82.5%
correct answers from retrieval: equal to handing the model the right file

5×
fewer prompt tokens than the right file (2.0k vs 10.1k)

8×
faster than pasting ~26k tokens of docs (8.5 s vs 69.8 s)

87%
of answers' source passages in the top 5 (EmbeddingGemma + reranker)

15×
faster reranking on the 6700 XT than the CPU (1 s vs 14 s)

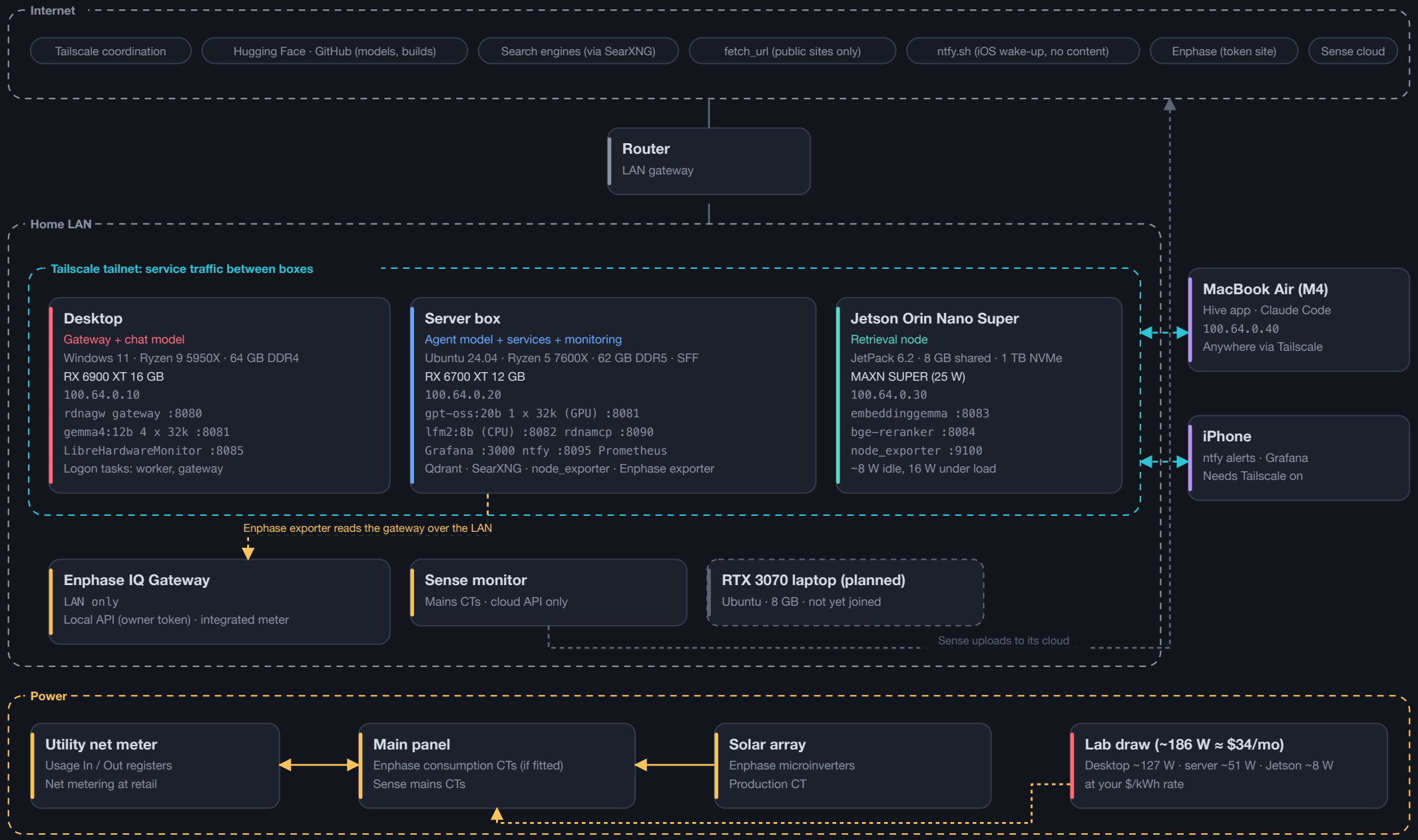
0 → 9/11
current-version questions right once gemma4 can search the web

0.2 s
titles and classification on the CPU model (LFM2), no GPU slot used

81 °C
6700 XT junction under load; 14 alert rules across both boxes, throttle at 110 °C

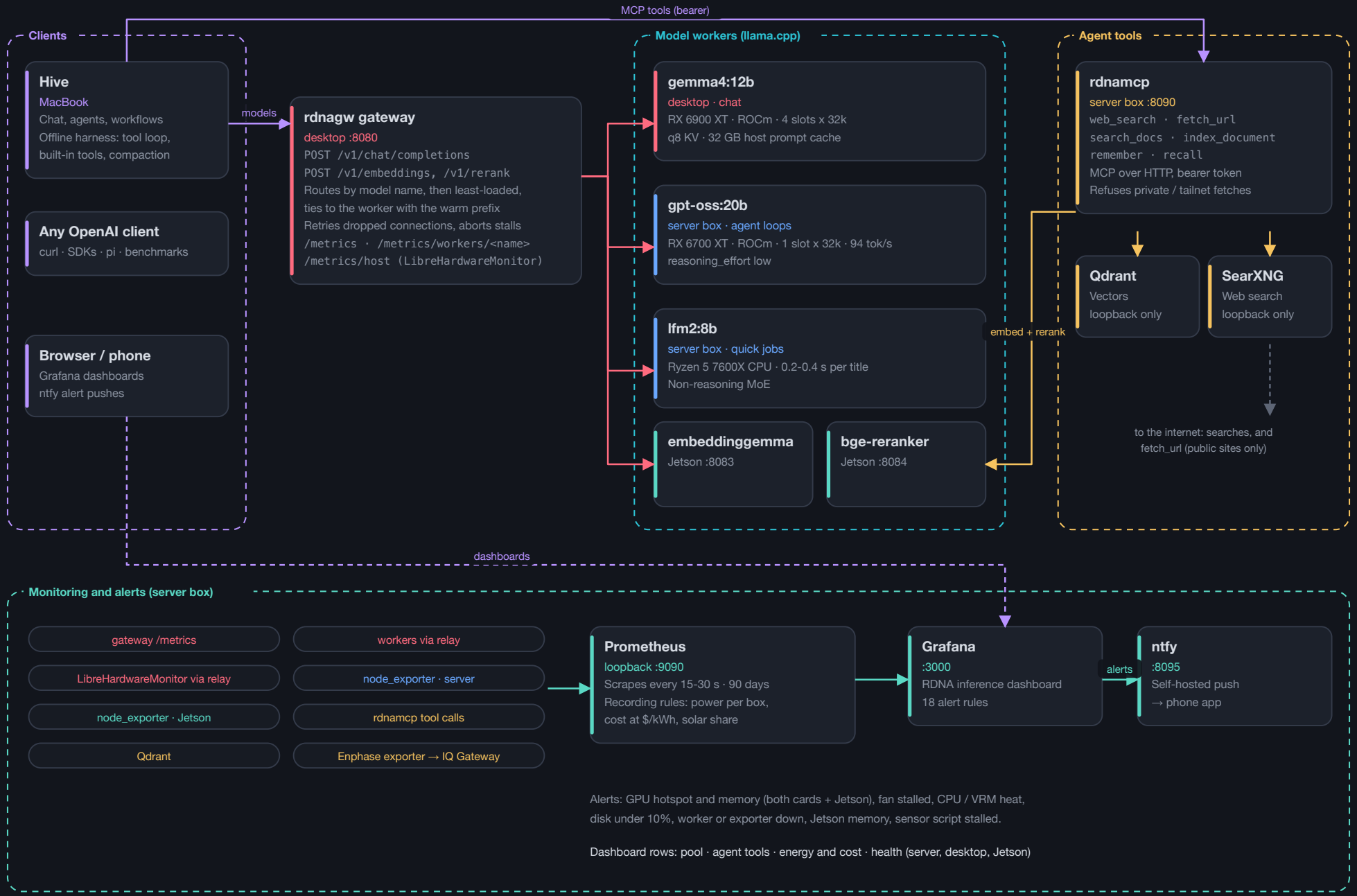
Homelab topology

Devices, networks and addresses. Service traffic rides Tailscale; only the solar gateway is LAN-only.



Service architecture

One OpenAI-compatible endpoint for models, one MCP endpoint for tools, Prometheus + Grafana watching both.



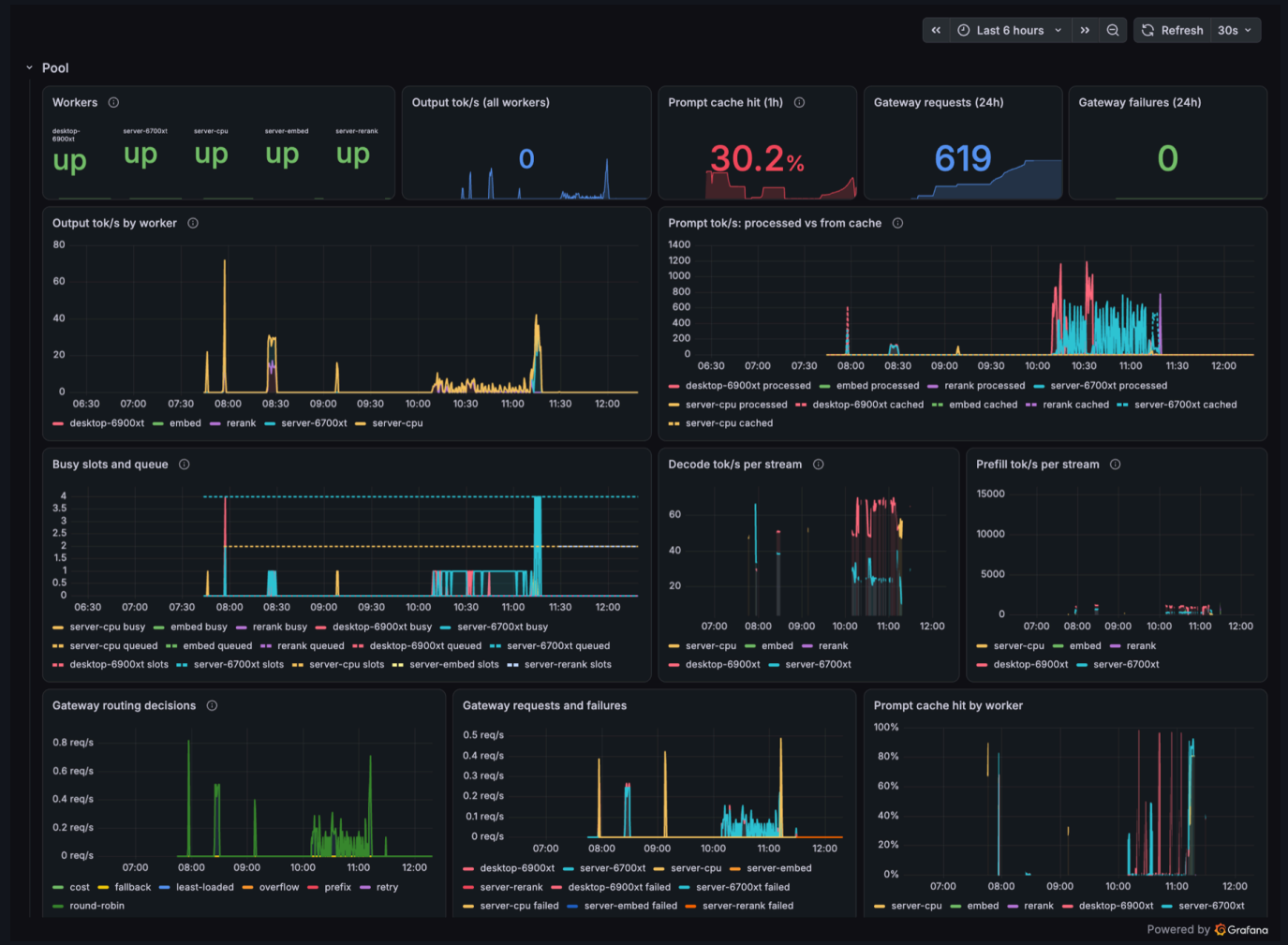
What runs on the server box

service	where	listens on
gemma4 12B worker (4×32k)	GPU, systemd	Tailscale :8081
EmbeddingGemma 300M · bge-reranker-v2-m3	GPU, systemd (`install.sh aux`)	Tailscale :8083 / :8084
LFM2 8B-A1B, quick jobs	CPU, container	Tailscale :8082
`rdnamcp`: 7 agent tools over MCP	container, bearer token	Tailscale :8090
Qdrant · SearXNG	containers	loopback only
Prometheus · node_exporter	containers	loopback only
Grafana · ntfy alerts	containers	Tailscale :3000 / :8095

The gateway serves `gemma4:12b` and `lfm2:8b` side by side and relays the desktop worker's metrics.

VRAM: 10.7 of 12.3 GB.

Monitoring: the inference pool



Grafana on the server box, last 6 hours (today's benchmark runs). Every worker behind the gateway, its tok/s, cache hits, slots and routing.

Monitoring: agent tools



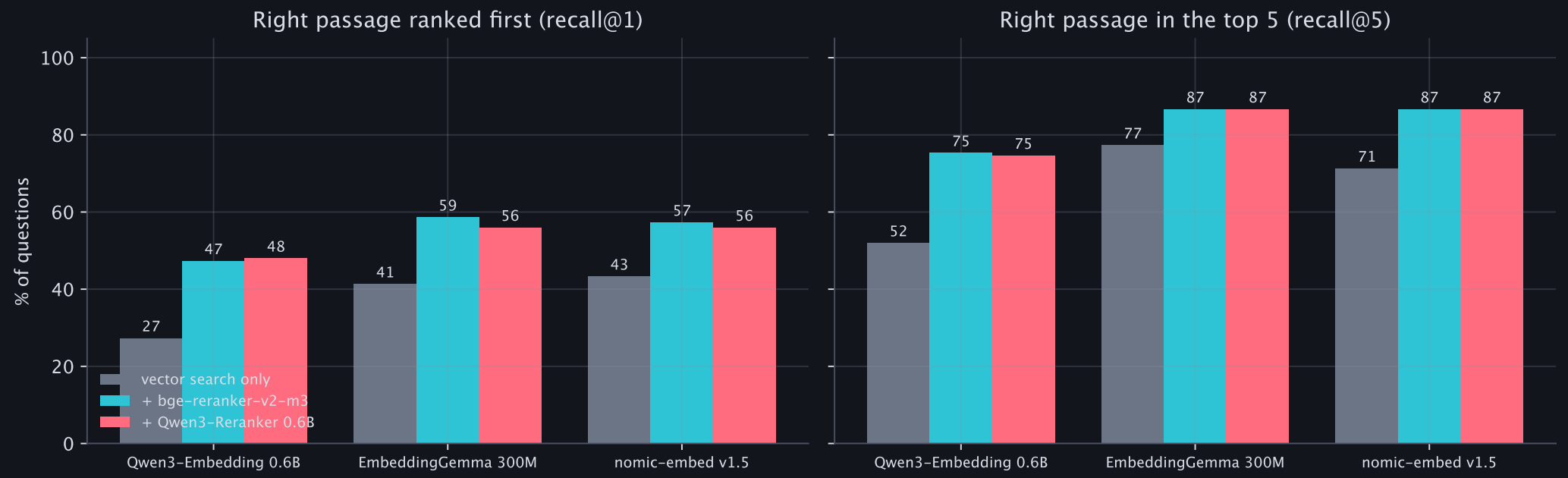
Calls and latency per MCP tool, tool errors (the one `fetch_url` error is a refused private address, by design), passages per collection, embed/rerank load and Qdrant latency.

Quick jobs belong on a non-reasoning model

CPU model (Ryzen 5 7600X)	decode	title · classify · pick an agent
LFM2 8B-A1B	52 tok/s	**0.2–0.4 s**, all correct
LFM2.5 8B-A1B	47 tok/s	~6 s: always reasons ~280 tokens first; can't be turned off

- 1B active parameters is fast enough on six Zen 4 cores, so short jobs never take a GPU slot.
- Not for research: with web tools it got only 4 of 11 version questions right, often inventing numbers.

Retrieval quality: rerank, and the smallest embedder wins



150 questions over llama.cpp's docs (1,732 passages); only the exact source passage counts.

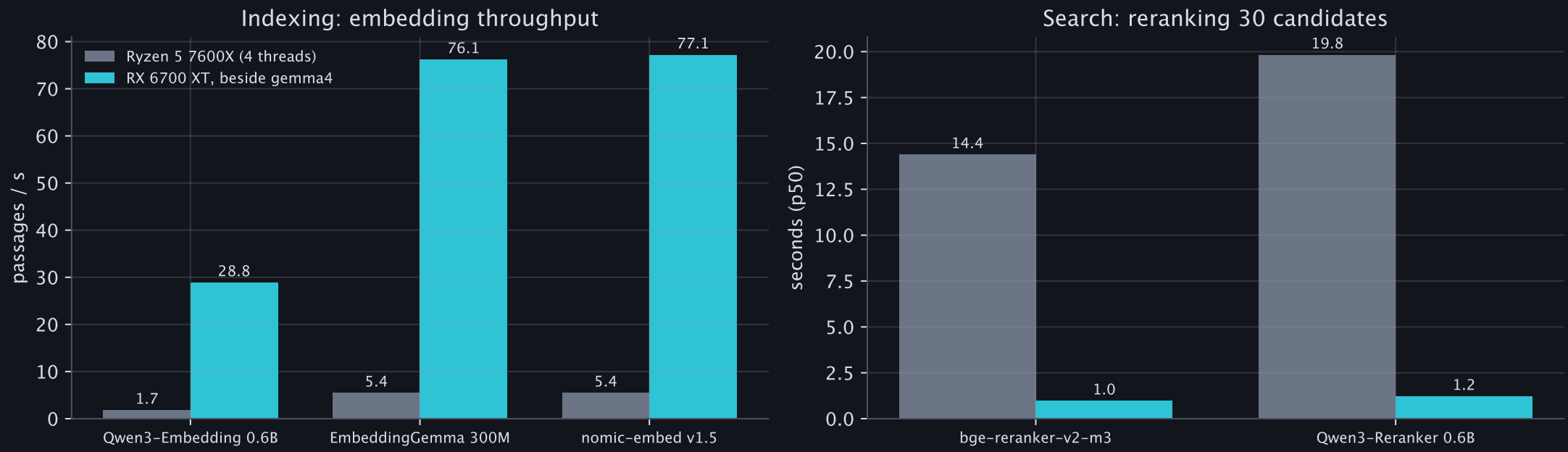
A reranker adds 10–15 points. EmbeddingGemma 300M beat Qwen3-Embedding 0.6B by 25 points before reranking.

How many candidates to rerank

EmbeddingGemma + bge-reranker	top 1	top 5	rerank time
vector search only	41%	77%	0
rerank top 10	56%	81%	~0.6 s
rerank top 20 (deployed)	**58%**	**85%**	**~1.2 s**
rerank top 30	59%	87%	1.8 s

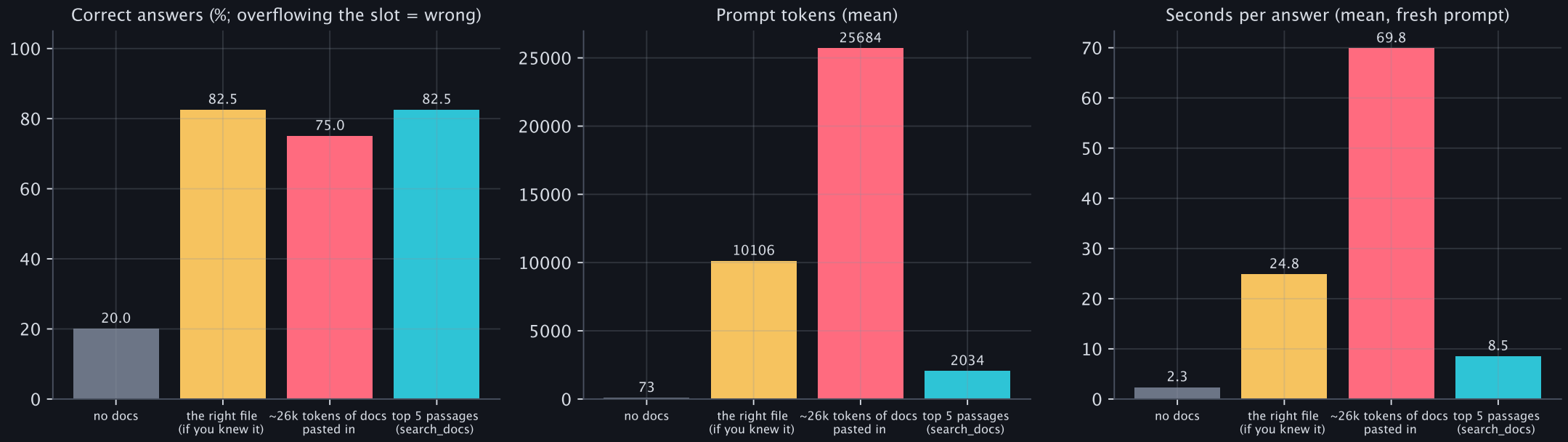
A cross-encoder scores each passage on its own, so one pass over 30 gives all three depths. Times are on the 6700 XT beside gemma4; 10 and 20 are scaled from 30.

Retrieval runs on the GPU



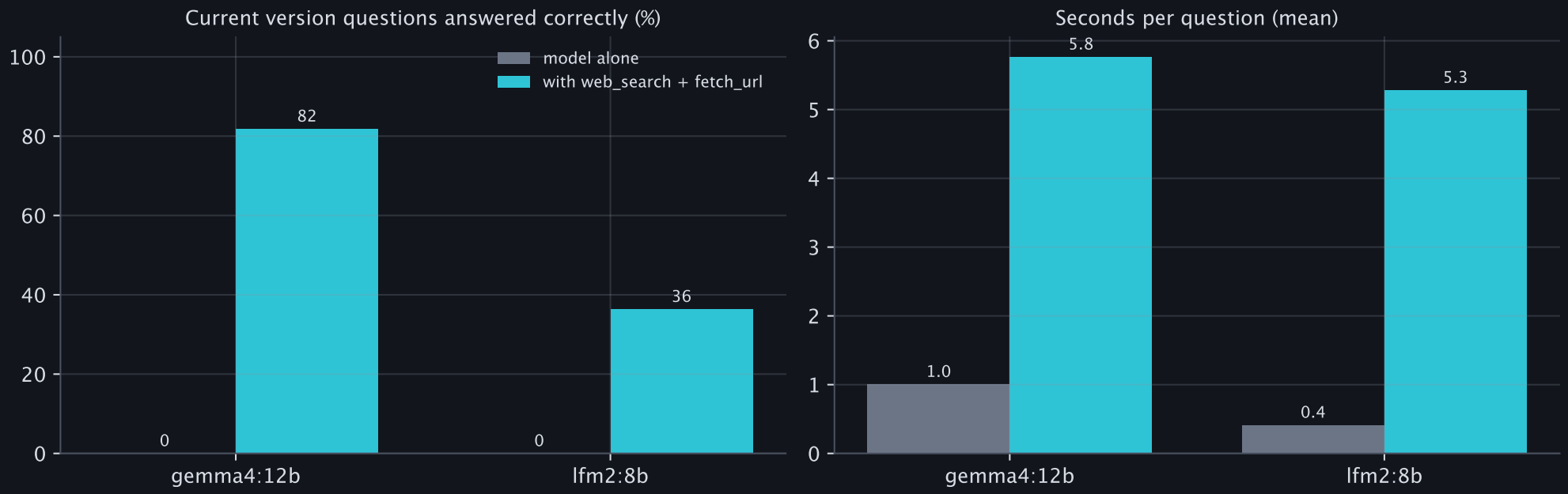
On the CPU a reranked search takes **14–20 s**; on the 6700 XT about **1–2 s**. Indexing is 15–17× faster. Both models fit beside the gemma4 worker with 1.5 GB to spare.

Retrieval vs. pasting docs into the prompt



Same accuracy as the right file, **a fifth of the tokens, a third of the time**. Pasted docs scored 97% when they fit, but **9 of 40 overflowed the 32k slot**: these docs run about 2.4 characters per token.

Web search: current facts



Live answers from go.dev, GitHub releases and endoflife.date. **gemma4 went from 0 to 9 of 11**, at 5.8 s per question; without tools it mostly declined. lfm2 searched, then still invented versions.

Contention: what retrieval costs the worker

background load (continuous)	gemma4 on the 6700 XT, 4 agents	lfm2 title job
none	39.6 / 49.7 tok/s (two runs)	0.23 s
CPU indexing	46.5 / 46.3 tok/s	0.28 s
GPU reranking	**24.9 tok/s**	0.22 s

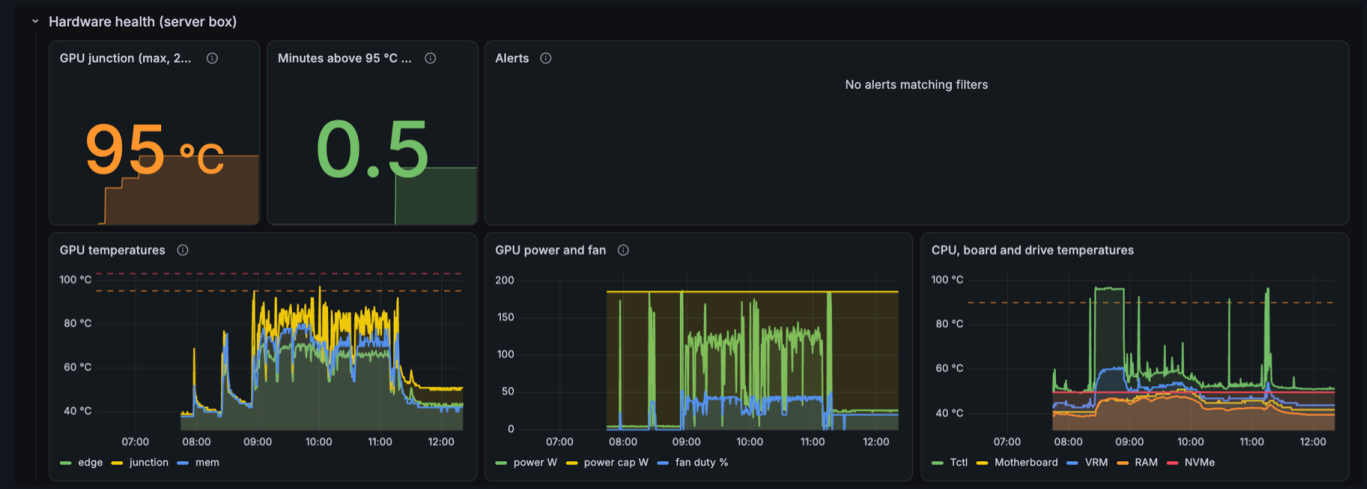
- CPU work doesn't touch the GPU worker.
- Back-to-back reranking halves generation on the shared card. A real search is a ~1 s burst, so occasional searches are cheap; steady search traffic would want its own GPU (the RTX 3070 laptop is a candidate).

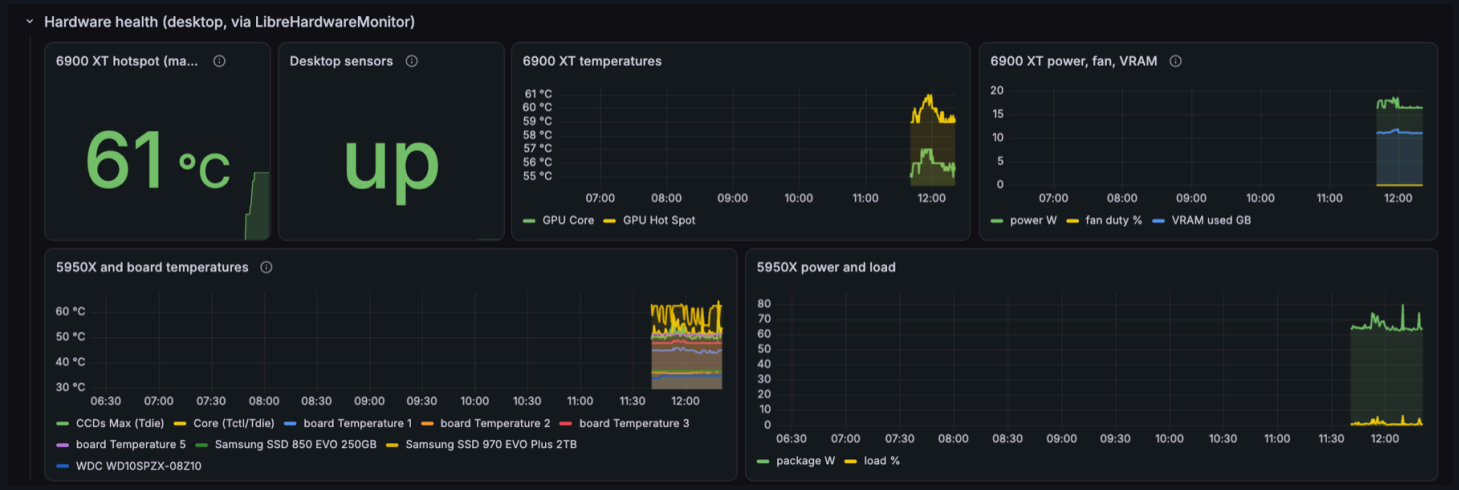
Hardware health and alerts

alert	threshold	card's own limit
GPU junction	warn > 95 °C for 10 min · critical > 103 °C for 2 min	throttles at 110 °C
GPU memory (GDDR6)	warn > 92 °C · critical > 100 °C	rated to 105 °C
GPU fan	0 rpm while drawing > 60 W	–
CPU · VRM	> 90 °C · > 100 °C for 5 min	Ryzen 7000: 95 °C
availability	disk < 10% free · worker or exporter down 5 min	–

Under the benchmark load: **81 °C junction, 72 °C memory, 123 W of a 186 W cap**. The desktop gets the same hotspot, fan and CPU alerts through LibreHardwareMonitor. 14 rules in all, pushed to a self-hosted ntfy topic.

Hardware health on the dashboard

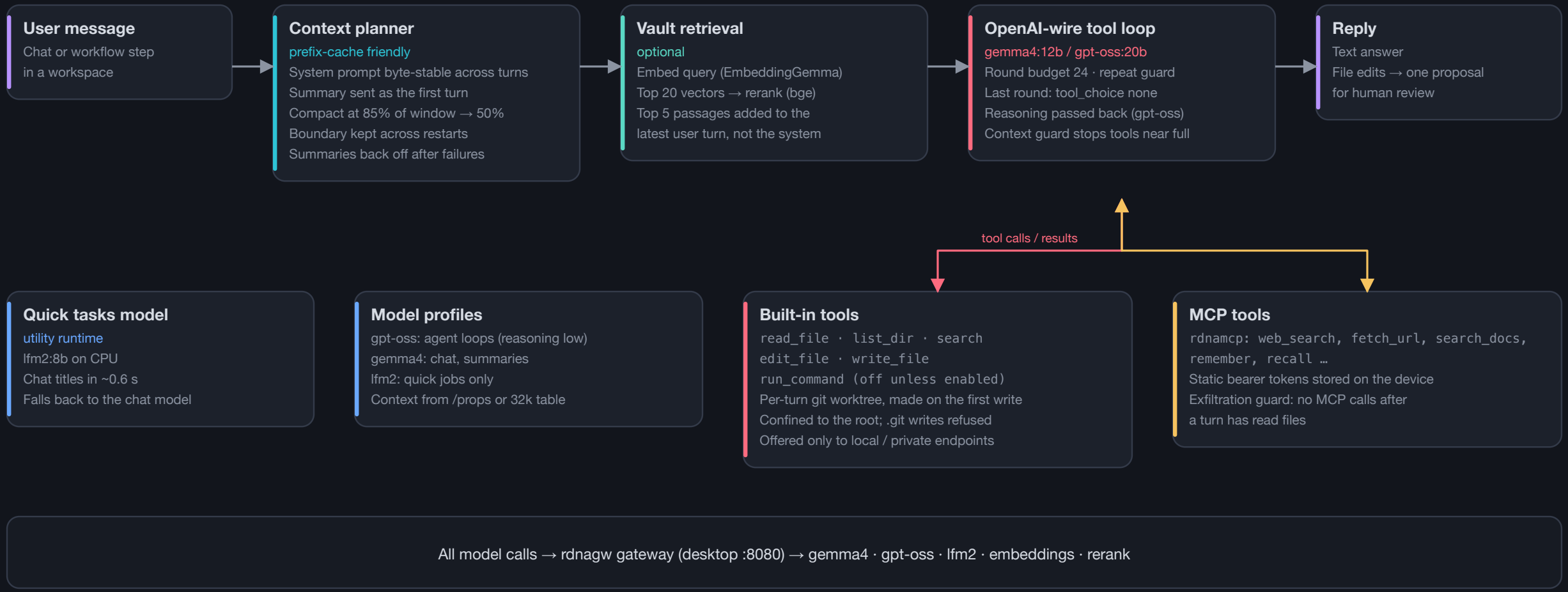




Top: 6700 XT through today's runs (95 °C peak, 30 s). Bottom: 6900 XT and 5950X via LibreHardwareMonitor.

Hive offline harness: one agent turn

Branch offline-harness. Every model call goes through the gateway; tools run locally or through rdnamcp.



Deployed configuration

piece	choice	why
embedder	EmbeddingGemma 300M, GPU	best recall, 5× faster indexing than Qwen3-Embedding
reranker	bge-reranker-v2-m3, GPU, top 20	ties Qwen3-Reranker, 25% faster; 85% top-5
chunks	1,200 characters, 200 overlap	
quick-jobs model	LFM2 8B-A1B, CPU	0.2–0.4 s, no GPU slot
research model	gemma4 12B + `web_search` / `fetch_url`	9/11 on live facts
fetch safety	refuse private, tailnet and local addresses	no path from an agent to Prometheus, k3s or the LAN
memory	`remember` / `recall`, shared by all agents	

Next steps for this lab

1. **Wire into Hive:** add the MCP server (HTTP, `:8090/mcp`, bearer token). Hive has no "small jobs" model yet: titles use the chat's own runtime, so `lfm2:8b` needs a new Hive setting (a utility runtime for titles and summaries) before it can take that work.
2. **Phone alerts:** subscribe the ntfy app to the alert topic (over Tailscale).
3. **RTX 3070 laptop** as a worker for the reranker and quick-jobs model, taking that load off the 6700 XT.
4. **Smaller chunks** (around 600 characters): long chunks dilute a sentence's score; top-1 is 59%.
5. **gpt-oss 20B on the 6700 XT** for agent loops: fits fully, 100 tok/s, kept 100% of facts through compaction.

Methodology and caveats

- Corpus: llama.cpp's Markdown docs at release b11723; questions written by gemma4 from random passages (seed 7).
- Strict metric: only the passage a question came from counts, though the docs repeat across backend guides.
- gemma4 graded the answers against the references; gemma4 wrote the questions, which may favour EmbeddingGemma slightly.
- Answering used fresh prompts (cache off), through the gateway, with both GPUs serving.
- Web truth fetched live at run time; the 11 questions change answers as releases happen.
- Contention loads were continuous worst cases; idle throughput varied between runs (39.6 vs 49.7 tok/s).

bench/rag.sh · cmd/ragbench · cmd/rdnamcp · deploy/sff · docs/BENCHMARKS.md

Main takeaways

1. **Retrieve, don't paste.** With 32k slots, the top 5 reranked passages answered as well as the right file, at a fifth of the tokens and 8× faster than pasted docs, which often overflow the slot.
2. **Run retrieval on the GPU, beside the chat model.** Embedder + reranker take ~1.5 GB of VRAM; on the CPU a reranked search costs 14–20 s. Rerank the top 20.
3. **Pick a small embedder, always rerank.** EmbeddingGemma 300M beat Qwen3-Embedding 0.6B by 25 points at 5× the indexing speed; a reranker adds 10–15.
4. **CPU for quick jobs, with a non-reasoning model.** LFM2 8B-A1B: titles and classification in 0.2–0.4 s on six cores. Reasoning models take 6 s; small models shouldn't do research.
5. **Give local models web tools.** gemma4 12B: 0 → 9 of 11 current facts. Block fetches to private and local addresses.
6. **Size for contention.** Back-to-back reranking halves generation on a shared card; bursty search is cheap, heavy search wants its own GPU.
7. **Alert on heat below the limits.** RDNA2 hotspots reach ~95 °C in sustained bursts; alert at 95 / 103 °C, under the 110 °C throttle.